FedAdaClip: Adaptive Clipping for Private Federated Learning

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***Abstract*—Privacy-preserving machine learning algorithms are essential in federated learning, enabling model training on user data while safeguarding sensitive information. Differential privacy federated learning algorithms, such as DP-FedAvg, address this by perturbing gradients with noise proportional to their sensitivity. However, key challenges remain, notably in selecting appropriate gradient clipping thresholds and adapting fixed values to heterogeneous data distributions. In this paper, we introduce `FedAdaClip`, a global gradient-aware adaptive clipping algorithm for differential privacy. `FedAdaClip` integrates online quantile estimation and gradient perception: during early training, quantile-based methods rapidly identify suitable threshold ranges; in later stages, thresholds are dynamically adjusted based on previous gradient norms, eliminating manual tuning. Theoretical analysis demonstrates that `FedAdaClip` minimizes composite clipping and noise errors under (ϵ, δ)-DP constraints, and guarantees convergence for non-convex objectives. Empirical results on MNIST and CIFAR-10/100 show that `FedAdaClip` outperforms fixed-threshold baselines such as DP-FedAvg and DP-SCAFFOLD, achieving 1.5%-7% higher test accuracy under equivalent privacy budgets. Moreover, training curves are smoother, substantially mitigating oscillations and divergence induced by suboptimal thresholds.**

***Index Terms*—federated learning, differential privacy, adaptive clipping**

I. Introduction

With the improvement of data privacy protection regulations such as GDPR and the Personal Information Protection Law, as well as the popularity of distributed application scenarios, federated learning (FL) [1] has become a key technology for solving the contradiction between data silos and privacy security. FL has demonstrated enormous application value in sensitive fields such as medical diagnosis, financial risk control, and IoT collaboration through the mode of local training on the client side and parameter federation aggregation. However, there remains a risk of privacy leakage in the parameter interaction process of FL. Attackers can reconstruct the original client data through gradient inversion, model inversion or membership inference [2]–[4].

The mainstream privacy protection methods currently applied to federated learning include homomorphic encryption (HE) [5], secure multi party computation (SMPC) [6] and differential privacy (DP) [7]. Compared to HE and SMPC, differential privacy can quantify privacy risks through parameters, with lower computational and communication overhead, and is more suitable for complex scenarios such as large-scale clients and non-IID data in federated learning. DP is typically enforced in FL through the DP-SGD algorithm [8], which operates by clipping the ℓ_2-norm of gradients to a fixed threshold C, followed by injecting Gaussian noise scaled by a noise parameter σ. The choice of the clipping threshold plays a pivotal role in balancing privacy and utility. A large threshold requires more noise and can degrade model accuracy, while a small threshold may truncate critical gradient information and hinder convergence [9].

The threshold clipping strategy in the current federated learning differential privacy mechanism has significant limitations and is difficult to adapt to the core challenges of practical application scenarios:

- **Biased gradient clipping.** At the data level, client data in federated learning commonly exhibits Non-Independent and Identically Distributed (Non-IID). The distribution discrepancies across different clients mean that fixed thresholds would cause excessive gradient clipping for some clients while leaving others inadequately clipped.
- **Suppressive convergence.** At the training level, the model convergence process demonstrates a dynamic nature. During early training stages, clients exhibit larger gradient norms, which gradually diminish in later stages. Fixed thresholds persistently suppress effective gradients in later phases, thereby slowing convergence speed.
- **Volatile aggregation.** At the aggregation level, excessively personalized local thresholds may cause oscillations in the federated model. The consistency of global parameter aggregation remains a significant challenge.

A. Our Results

In this paper, we address fundamental challenges in differentially private FL. We propose `FedAdaClip`—a novel adaptive gradient clipping mechanism designed specifically for differentially private federated learning. Our key contributions are summarized as follows:

- **Novel Perspective on Clipping Imbalance in Non-IID FL:** We are the first to systematically highlight the inherent tension between accuracy and privacy caused by the use of a *fixed* gradient clipping threshold in

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federated learning with non-IID data. To resolve this, we propose `FedAdaClip`, an adaptive clipping mechanism that leverages global gradient statistics to dynamically adjust the clipping threshold. This approach enables fully *online adaptation* without requiring public datasets or repeated hyperparameter tuning.

- **Theoretical Guarantees on Privacy, Optimality, and Convergence:** We develop a comprehensive theoretical analysis of `FedAdaClip`, demonstrating that its adaptive clipping strategy achieves near-optimal noise variance under differential privacy constraints. Specifically, we: Derive the analytical solution for the optimal adaptive parameters (a, b); Establish a tight upper bound on the total error introduced by noise and clipping under (ϵ, δ)-DP; Prove convergence guarantees for non-convex optimization objectives under standard assumptions.
- **Superior Empirical Performance on Benchmark Datasets:** Through extensive experiments on MNIST, CIFAR-10, and CIFAR-100, we show that `FedAdaClip` consistently outperforms state-of-the-art baselines including DP-FedAvg and DP-SCAFFOLD. Our method improves test accuracy by **1.5%–7.0%**, delivers smoother and more stable training curves, and avoids divergence issues arising from misconfigured clipping thresholds.

B. Related Work

In differentially private federated learning (DP-FL), the choice of the gradient clipping threshold is a critical factor that directly impacts both model accuracy and privacy budget consumption. Existing research in this domain primarily follows three directions: (i) fixed threshold methods, (ii) public data-assisted strategies, and (iii) heuristic adaptive approaches. However, none of these methods simultaneously satisfy the three key requirements of *no public data dependency*, *online adaptability*, and *theoretical convergence guarantees*.

- **Fixed Threshold Methods.** Representative works such as DP-FedAvg [10] and DP-SCAFFOLD [11] adopt a fixed clipping threshold C as originally proposed by [8]. This threshold remains constant throughout training. However, in non-IID federated settings, this strategy often leads to suboptimal performance—either over-clipping informative gradients or introducing excessive noise—thereby significantly degrading model accuracy [9].
- **Public Data-Assisted Methods.** To better calibrate the clipping threshold, several approaches rely on access to auxiliary public data. For instance, Zhang *et al.* [12] use public datasets to estimate an upper bound for the gradient norm, while Geyer *et al.* [13] assume that the server has access to additional data to adjust the noise scale. However, in most realistic federated learning scenarios—especially in privacy-sensitive applications—sharing or relying on public data is infeasible or prohibited.
- **Single-Machine Adaptive Methods.** Adaptive clipping strategies such as AdaClip [14] attempt to dynamically adjust the threshold using exponential moving averages of past gradients. However, AdaClip lacks robust statistical mechanisms (e.g., quantile estimation), resulting in high threshold volatility. Conversely, FedMAC [15] relies solely on quantile tracking without incorporating local gradient dynamics, which often leads to slower early-stage convergence. Moreover, neither approach provides a unified theoretical analysis of both differential privacy and convergence guarantees.

In conclusion, existing works fail to simultaneously address the core requirements: (i) no reliance on public data, (ii) fully online threshold adaptation, and (iii) provable convergence under differential privacy constraints. To bridge this gap, our proposed `FedAdaClip` mechanism incorporates quantile-based clipping and gradient-aware dynamics within a unified adaptive framework. Notably, it is complemented by rigorous theoretical analysis, which establishes both (ϵ, δ)-DP guarantees and convergence results under non-convex objectives.

II. Preliminary

In this section, we introduce the foundational concepts that underpin our proposed method, including differentially private gradient perturbation, clipping strategies, and notations commonly used in federated learning.

A. Federated Learning and Notation

Federated learning aims to collaboratively train a global model across multiple clients without sharing their raw data. Let $\mathcal{D}_i$ denote the local dataset on client i, and $f_i(\cdot)$ be the corresponding local objective. The global optimization problem is [16]:

$$\min_{\boldsymbol{x}} F(\boldsymbol{x}) := \sum_{i=1}^{N} p_i f_i(\boldsymbol{x}), \tag{1}$$

where $p_i = \frac{|\mathcal{D}_i|}{\sum_j |\mathcal{D}_j|}$ denotes the data proportion on client i. At each round t, a subset of clients $\mathcal{S}_t$ is sampled to perform K local updates, followed by model aggregation on the server.

B. Differential Privacy in FL

To protect sensitive client information, differential privacy is commonly employed in FL via gradient perturbation. In a typical setup, local gradients are clipped to bound sensitivity and then injected with Gaussian noise [17]:

$$\tilde{\boldsymbol{g}}_i = \text{clip}(\boldsymbol{g}_i, C) + \mathcal{N}(0, \sigma^2 C^2 I), \tag{2}$$

where C is the clipping threshold, σ controls the noise magnitude, and I is the identity matrix. The privacy budget ϵ is then analyzed via standard composition theorems.

C. Bias and Variance of Gradient Clipping

During the t-th iteration on the server, consider the k-th local update performed by the i-th client, with the gradient denoted as $\boldsymbol{g}^t = (g_1^t, g_2^t, \ldots, g_d^t)$. For simplicity, the superscript k and subscript i are omitted here. Additionally, two auxiliary vectors $\boldsymbol{a}^t = (a_1^t, a_2^t, \ldots, a_d^t)$ and $\boldsymbol{b}^t = (b_1^t, b_2^t, \ldots, b_d^t)$ are introduced, both having the same dimension as $\boldsymbol{g}^t$. Define

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$\boldsymbol{\tau}^t = \frac{\boldsymbol{g}^t - \boldsymbol{a}^t}{\boldsymbol{b}^t}$, where each element $\boldsymbol{\tau}_j^t = \frac{\boldsymbol{g}_j^t - \boldsymbol{a}_j^t}{\boldsymbol{b}_j^t}$ represents the result of subtracting each component of $\boldsymbol{a}^t$ from $\boldsymbol{g}^t$ and then dividing by the corresponding component of $\boldsymbol{b}^t$. To constrain the sensitivity, the transformed gradient $\boldsymbol{\tau}^t$ is clipped to its ℓ_2 norm [14]:

$$\hat{\boldsymbol{\tau}}^t = \operatorname{clip}\left(\boldsymbol{\tau}^t, 1\right) \triangleq \frac{\boldsymbol{\tau}^t}{\max\left(1, \|\boldsymbol{\tau}^t\|_2^2\right)}. \tag{3}$$

Then add noise to the gradient $\hat{\boldsymbol{\tau}}^t$after threshold clipping:

$$\tilde{\boldsymbol{\tau}}^t = \hat{\boldsymbol{\tau}}^t + \mathcal{N}^t\left(0, \sigma^2 C^2 \boldsymbol{I}_m\right). \tag{4}$$

In differentially private federated learning, to protect privacy, noise is typically added to gradient updates. The standard deviation σ of the noise is determined by the privacy parameter ϵ. Specifically, the gradient after clipping and adding noise, $\widetilde{\boldsymbol{g}}^t$, can be expressed as:

$$\tilde{\boldsymbol{g}}^t = \boldsymbol{b}^t \widetilde{\boldsymbol{\tau}}^t + \boldsymbol{a}^t. \tag{5}$$

Here, $\widetilde{\boldsymbol{\tau}}^t$ represents the gradient after clipping and adding noise, while $\boldsymbol{a}^t$ and $\boldsymbol{b}^t$ are auxiliary vectors. By applying the triangle inequality and Jensen's inequality, we obtain the following upper bounds on bias and variance:

$$\begin{aligned} \mathbf{D}\left(\tilde{\boldsymbol{g}}^t\right) &\leq \mathbf{D}\left(\boldsymbol{g}^t\right) + 2\mathbb{E}\left\|\tilde{\boldsymbol{g}}^t - \boldsymbol{g}^t\right\|, \\ \mathbf{V}\left(\tilde{\boldsymbol{g}}^t\right) &\leq 3\mathbf{V}\left(\boldsymbol{g}^t\right) + 6\mathbb{E}\left\|\tilde{\boldsymbol{g}}^t - \boldsymbol{g}^t\right\|^2, \end{aligned} \tag{6}$$

where $\mathbf{D}(\boldsymbol{x}) \triangleq \|\mathbb{E}\boldsymbol{x} - \boldsymbol{x}\|$ denotes the bias and $\mathbf{V}(\boldsymbol{x}) \triangleq \mathbb{E}\|\boldsymbol{x} - \mathbb{E}\boldsymbol{x}\|^2$ denotes the variance. According to Theorem 1, to find the optimal $\boldsymbol{a}^t$ and $\boldsymbol{b}^t$, it is necessary to constrain $\mathbb{E}\left\|\tilde{\boldsymbol{g}}^t - \boldsymbol{g}^t\right\|$. Further analysis yields:

$$\mathbb{E}\left\|\widetilde{\boldsymbol{g}}^t - \boldsymbol{g}^t\right\|^2 = \underbrace{\left\|\boldsymbol{g}^t - \boldsymbol{a}^t\right\|\left(1 - \frac{1}{\max(1, \|\boldsymbol{\tau}^t\|_2^2)}\right)}_{A_t} + \underbrace{\left\|\boldsymbol{b}^t\right\|^2 \sigma^2}_{B_t}. \tag{7}$$

The first term A_t on the right-hand side of the equation represents the bias introduced by the clipping operation, while the second term B_t represents the error caused by the added noise. Therefore, minimizing $\mathbb{E}\left\|\tilde{\boldsymbol{g}}^t - \boldsymbol{g}^t\right\|^2$ can be transformed into optimizing the clipping threshold. This transformation provides a clearer perspective, making the essence of the problem easier to understand and solve. Based on this, we propose an adaptive strategy for dynamically adjusting the clipping threshold to bring $\widetilde{\boldsymbol{\tau}}^t$ closer to the target quantile.

D. Quantile-Based Clipping

Consider a d-dimensional random variable $\boldsymbol{x} \in \mathbb{R}^d$, and let $\gamma \in [0, 1]$ be the quantile to be empirically determined. Given an arbitrary clipping threshold C, the function is defined as follows [18]:

$$\ell_\gamma(C; \boldsymbol{x}) = \begin{cases} (1-\gamma)(C - \boldsymbol{x}), & \text{if } \boldsymbol{x} \leq C \\ \gamma(\boldsymbol{x} - C), & \text{otherwise} \end{cases} \tag{8}$$

Taking the derivative of the above function, we obtain:

$$\ell'_\gamma(C; \boldsymbol{x}) = \begin{cases} 1 - \gamma, & \text{if } \boldsymbol{x} \leq C \\ -\gamma, & \text{otherwise} \end{cases} \tag{9}$$

It follows that $\mathbb{E}\left[\ell'_\gamma(C; \boldsymbol{x})\right] = (1-\gamma)\Pr[\boldsymbol{x} \leq C] - \gamma\Pr[\boldsymbol{x} > C] = \Pr[\boldsymbol{x} \leq C] - \gamma$. Let C^* be the optimal clipping threshold such that $\mathbb{E}\left[\ell'_\gamma\left(C^*; \boldsymbol{x}\right)\right] = 0$, which implies $\Pr\left(\boldsymbol{x} \leq C^*\right) = \gamma$, i.e., C^* is the γ-th quantile of the random variable $\boldsymbol{x}$. According to Theorem 2 in [15], the clipping threshold C obtained by gradient descent will converge to the γ-th quantile of the random variable $\boldsymbol{x}$. Assuming the components of the random variable $\boldsymbol{x}$ are represented as $(x_1, x_2, \ldots, x_d)$, further analysis of its properties can be conducted.

E. Gradient-Aware Clipping

In practical federated learning applications, significant heterogeneity arises from factors such as data distribution, dataset size, and data sampling methods of different clients. This heterogeneity makes a single fixed clipping threshold strategy insufficient. To better adapt to this heterogeneity, we propose a gradient-aware adaptive clipping strategy. This strategy dynamically adjusts the clipping threshold by multiplying the l_2 norm of the differentially private gradient from the previous iteration by a constant factor ρ. The specific formula for the clipping threshold C_t is as follows:

$$C_t \leftarrow \rho C_{t-1} \cdot \exp(-\eta_l(\boldsymbol{b} - \gamma)) + (1 - \rho)\left(\frac{1}{S}\sum_{i \in S^t} \|\boldsymbol{g}\|_2^2\right), \tag{10}$$

where $\operatorname{clip}\left(\|\cdot\|_2\right) = \frac{\|\cdot\|_2}{\max(1, \|\cdot\|_2 / C_{t-1})}$.

This gradient-aware clipping strategy can dynamically adjust the clipping threshold based on the gradient of each client, thus better adapting to the heterogeneity among different clients.

III. Model

In this section, we introduce the system model and threat model of FedAdaClip.

A. System Model

We consider a standard federated learning framework composed of a central server and N distributed clients, as shown in Fig 1. The objective is to collaboratively optimize a global model while ensuring that sensitive local data on each client remains private.

The training process is conducted in iterative communication rounds. At the beginning of training, the server initializes the global model parameters and global gradient, then broadcasts them to all clients. In each round, every client computes the gradient of its local loss function using its private dataset. To protect privacy and enhance robustness under heterogeneous data, FedAdaClip introduces a two-stage adaptive clipping strategy. Specifically, quantile estimation is first employed to rapidly determine a reasonable clipping

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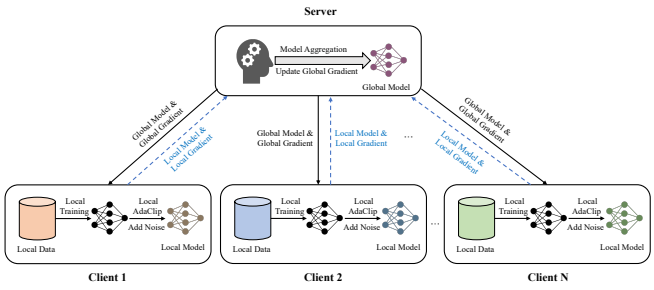


Fig. 1: System framework of FedAdaClip

threshold range, and in subsequent rounds the threshold is dynamically adjusted according to the previous gradient norms. This mechanism mitigates the bias introduced by manually fixed thresholds and adapts to non-IID data distributions. After clipping, each client adds Gaussian noise calibrated to the sensitivity of the clipped gradients to guarantee (ϵ, δ)-DP. The noisy gradients and local model are then transmitted to the server, which aggregates them to update the global model and global gradient. This iterative process continues until convergence, after which the server outputs the final privacy-preserving global model.

B. Threat Model

For the threat model, we assume a semi-honest adversary, which could be either the central server or a subset of colluding clients among the N participants. The adversary strictly follows the federated learning protocol but attempts to infer sensitive information from the uploaded gradients or global model parameters. Potential privacy leaks include the revelation of specific features, labels, or the presence of individual samples in a client's local dataset. The adversary may employ attacks such as gradient inversion or model extraction to exploit these information sources.

The FedAdaClip scheme is designed to mitigate such risks by ensuring that, under (ϵ, δ)-DP constraints, the adversary cannot distinguish between two datasets that differ by a single sample. Additionally, the adaptive clipping strategy minimizes the composite error from clipping and noise injection, balancing privacy protection with model performance even when dealing with heterogeneous data across the N clients. This defense mechanism ensures robustness against the identified threats while maintaining convergence guarantees for non-convex objectives.

IV. Our Scheme

Based on these discussions, we propose a new adaptive clipping algorithm named `FedAdaClip` for private federated learning, as shown in Algorithm 1. The algorithm combines exponential moving average (EMA) techniques to integrate quantile-based clipping and gradient-aware strategies, aiming to improve the precision and adaptability of gradient clipping in federated learning. The core idea of the `FedAdaClip` algorithm is to use the quantile-based clipping strategy in the early stages of training to quickly stabilize the clipping threshold and avoid oscillations caused by improper threshold settings. In the later stages of training, the gradient-aware strategy dynamically adjusts the clipping threshold to adapt to the heterogeneity of different client data, thus enhancing the overall performance of the model.

Algorithm 1 FedAdaClip

Input: Server $\{\boldsymbol{x}^0, \boldsymbol{g}^0, \eta_g\}$, Clients $\{\boldsymbol{g}^0, \eta_l\}$, EMA decay rates ρ, experience score $\boldsymbol{b}$, target quantile γ, momentum coefficient β

Output: Final model $\boldsymbol{x}^T$

1: Initialize $b_0 = \gamma$, $\boldsymbol{g}^0 = 0$
2: **for** $t = 0, \ldots, T-1$ **do**
3: Sample client subset $S^t \subset [N]$ of size $S = \lfloor lN \rfloor$
4: Server broadcasts $(\boldsymbol{x}^t, \boldsymbol{g}^t)$ to clients in S^t
5: **for** each client $i \in S^t$ **in parallel do**
6: Initialize local model: $\boldsymbol{x}_i^0 \leftarrow \boldsymbol{x}^t$, $\boldsymbol{g}^t \leftarrow \boldsymbol{g}^t$
7: **for** $k = 0, \ldots, K-1$ **do**
8: Sample mini-batch $S_i^k \subset D_i$ of size $\lfloor \alpha S \rfloor$
9: Compute gradient $\boldsymbol{g}_i(\boldsymbol{x}_i^k)$
10: Clip gradient: $\tilde{\boldsymbol{g}}_i \leftarrow \boldsymbol{g}_i / \max(1, \|\boldsymbol{g}_i\|_2 / C_t)$
11: $\boldsymbol{H}_i^k \leftarrow \frac{1}{sR} \sum_{j=1}^{R} \boldsymbol{g}_j + \frac{2C_t}{sR} \mathcal{N}(0, \sigma^2 C_t^2 I)$
12: $\boldsymbol{x}_i^{k+1} \leftarrow \boldsymbol{x}_i^k - \eta_l((1-\beta)\boldsymbol{H}_i^k + \beta \boldsymbol{g}^t)$
13: **end for**
14: Send local changes to server: $\Delta \boldsymbol{x}_i^t \leftarrow \boldsymbol{x}_i^0 - \boldsymbol{x}_i^K$
15: **end for**
16: Aggregate updates: $\Delta \boldsymbol{x}^t \leftarrow \frac{1}{|S^t|} \sum_{i \in S^t} \Delta \boldsymbol{x}_i^t$
17: Server updates: $\boldsymbol{x}^{t+1} \leftarrow \boldsymbol{x}^t + \eta_g \Delta \boldsymbol{x}^t$,
18: $\boldsymbol{g}_i = \frac{1}{\eta K} \Delta \boldsymbol{x}^t$, $\boldsymbol{g} = \frac{1}{|S^t|} \sum_{i \in S^t} \boldsymbol{g}_i$
19: Update clipping threshold: $C_t \leftarrow \rho C_{t-1} \cdot \exp(-\eta_l(\boldsymbol{b} - \gamma)) + (1-\rho)\left(\frac{1}{S} \sum_{i \in S^t} \|\boldsymbol{g}\|_2^2\right)$
20: Update experience score: $\boldsymbol{b} \leftarrow \frac{1}{S} \sum_{i=1}^{S} \mathbb{I}_{\|\boldsymbol{g}\| \le C_t}$
21: **end for**

The server initializes global model parameters and sets training hyperparameters (lines 1–2). For each iteration t, the server randomly samples client subset $S^t \subset [N]$ of size $S = \lfloor lN \rfloor$ and broadcasts $(\boldsymbol{x}^t, \boldsymbol{g}^t)$ to clients in S^t (lines 3–4).

In each local training iteration (lines 5–15), client i first samples a mini-batch $S_i^k \subset D_i$ of size $\lfloor \alpha S \rfloor$. Then, the client computes the corresponding gradient $\boldsymbol{g}_i(\boldsymbol{x}_i^k)$ and applies gradient clipping to bound its norm:

$$\tilde{\boldsymbol{g}}_i \leftarrow \frac{\boldsymbol{g}_i}{\max\left(1, \|\boldsymbol{g}_i\|_2 / C_t\right)}, \tag{11}$$

where C_t is the global clipping parameter.

Next, the client obtains a privatized estimate by averaging over R gradient samples and injecting Gaussian noise to ensure differential privacy:

$$H_i^k \leftarrow \frac{1}{sR} \sum_{j=1}^{R} \boldsymbol{g}_j + \frac{2C_t}{sR} \mathcal{N}(0, \sigma^2 C_t^2 I), \tag{12}$$

where $\mathcal{N}\left(0, \sigma^2 C_t \boldsymbol{I}_m\right)$ is the added Gaussian noise.

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Finally, the local model is updated using the learning rate η_l:

$$\boldsymbol{x}_i^{k+1} \leftarrow \boldsymbol{x}_i^k - \eta_l((1-\beta)H_i^k + \beta \boldsymbol{g}^t), \tag{13}$$

where β is the momentum coefficient and $\boldsymbol{g}^t$ denotes the global correction term.

On the server side (lines 16–20), after receiving the local updates from the sampled clients S^t, the global model is updated as

$$\boldsymbol{x}^{t+1} \leftarrow \boldsymbol{x}^t + \eta_g \Delta \boldsymbol{x}_t, \tag{14}$$

where η_g is the server learning rate and $\Delta \boldsymbol{x}_t$ denotes the aggregated client update at round t.

Server update is normalized as

$$\boldsymbol{g}_i = \frac{1}{\eta K}\Delta \boldsymbol{x}_i^t, \boldsymbol{g} = \frac{1}{|S^t|}\sum_{i \in S^t} \boldsymbol{g}_i, \tag{15}$$

where η is the local learning rate and K is the number of local steps. DP-FedAvg and AdaClip may cause significant deviations between local updates and the global model, leading to unstable convergence. In contrast, `FedAdaClip` aligns local and global update trajectories through a global clipping mechanism, effectively reducing model oscillations.

The server then updates the global clipping threshold C_t by combining an exponential moving average term with a gradient-aware adjustment:

$$C_t \leftarrow \rho C_{t-1} \cdot \exp(-\eta_l(\boldsymbol{b} - \gamma)) + (1-\rho)\left(\frac{1}{S}\sum_{i \in S^t} \|\boldsymbol{g}\|_2^2\right), \tag{16}$$

where ρ is the EMA decay parameter, η_l is the local learning rate, $\boldsymbol{b}$ is the experience score, and γ is a target threshold.

Finally, the experience score $\boldsymbol{b}$ is updated according to the proportion of client updates that lie within the clipping threshold:

$$\boldsymbol{b} \leftarrow \frac{1}{S}\sum_{i=1}^{S} \mathbb{I}_{\|\boldsymbol{g}\| \le C_t}. \tag{17}$$

This adaptive strategy not only quickly locates the appropriate clipping threshold in the early stages of training but also dynamically adjusts according to gradient changes in the later stages, thereby improving the convergence speed and accuracy of the model while protecting privacy.

V. Theoretical Analysis

This section provides a theoretical analysis of the `FedAdaClip` mechanism, focusing on its noise minimization and convergence properties. First, we detail the noise minimization mechanism of `FedAdaClip` and formally prove that the gradient updates under this framework strictly satisfy differential privacy, ensuring effective privacy protection for participants in federated learning. Second, we analyze the convergence of `FedAdaClip`, demonstrating its stable and rapid convergence to the global optimum during the iterative process. This not only ensures good training performance with minimized noise but also supports the algorithm's feasibility in practical applications.

A. Differential Privacy Bounds

Input perturbation effectively protects the privacy of the original data but severely compromises data utility by adding noise to the input data [22], which leads to a significant drop in model performance. On the other hand, objective perturbation provides privacy protection by adding noise to the objective function [23]. However, perturbing the objective function may prevent the model from converging to the optimum and affect model performance. The last approach is gradient perturbation [24], which adds noise to the gradients of the objective function during the gradient descent process. Thus, it satisfies differential privacy while still obtaining relatively optimal models. Since input perturbation and objective perturbation cannot converge to the global optimum, they are not applicable when dealing with non-convex loss functions [25]. Therefore, gradient perturbation becomes the most widely used approach.

Definition 1 (Neighboring datasets [7]). *Let $\mathcal{D}^n$ be the domain of all datasets, which can be viewed as any datasets. $D, D' \in \mathcal{D}^n$ are called neighboring if and only if D, D' differ in one element $i.e.$, D and D' can be obtained from each other by adding or removing an element.*

Next, we provide the common definition of differential privacy and Rényi differential privacy.

Definition 2 ((ϵ,δ)-DP [26]). *A randomized mechanism $\mathcal{M} : x^n \to \mathcal{Y}$ satisfies (ϵ,δ)-DP if for any pairs of neighboring datasets $D, D' \in \mathcal{D}$, and any output subset $O \subseteq \mathcal{Y}$, it has $\Pr[\mathcal{M}(D) \in O] \le e^\epsilon \Pr[\mathcal{M}(D') \in O] + \delta$.*

Definition 3 ((α,ϵ)-RDP [27]). *For any $\alpha > 1$ and any $\epsilon > 0$, a randomized mechanism $\mathcal{M} : x^n \to \mathcal{Y}$ satisfies (α,ϵ)-RDP, if for all neighboring datasets $D, D' \in \mathcal{D}$, $D_\alpha\big(\mathcal{M}(D)||\mathcal{M}(D')\big) := \frac{1}{\alpha-1}\log E\big(\frac{\mathcal{M}(D)}{\mathcal{M}(D')}\big)^\alpha \le \epsilon$.*

Before giving the lemma of conversion from RDP to DP, the concrete form of RDP under the Gaussian mechanism should be given in advance.

Lemma 1 (RDP Gaussian mechanism [27]). *If $f : x^n \to \mathbb{R}^d$ has ℓ_2-sensitivity, then the Gaussian mechanism $G_f(\cdot) := f(\cdot) + \mathcal{N}(0, \sigma^2 I_d)$ is $(\alpha, \alpha/2\sigma^2)$-RDP for any $\alpha > 1$.*

Lemma 2 (RDP to DP conversion [27]). *If a randomized mechanism $\mathcal{M}$ satisfies (ϵ,δ)-RDP, then $\mathcal{M}$ is $(\epsilon + \log(1/\delta)/(\alpha-1), \delta)$-DP for any $\delta \in (0,1)$.*

It is worth noting that we mainly refer to the user-level DP under uniformly subsampling clients, *i.e.*, subsampling clients of size $|\boldsymbol{S}_t|$ in each iteration t.

Theorem 1 (Privacy Bounds). *$\forall \epsilon, \delta \in (0,1)$, the output $\boldsymbol{x}^t$ is $(O(\epsilon), \delta)$-DP in user-level for `FedAdaClip` with uniformly subsampling when $N(0, \sigma^2 I)$ satisfies:*

$$\sigma \ge \frac{\tau \mathcal{L}}{\epsilon}\sqrt{\frac{14T}{\lambda}\left(\frac{\log(1/\delta)}{1-\lambda} + \epsilon\right)}, \tag{18}$$

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TABLE I: Comparison between FedAdaClip with other DP-FL schemes.

Scheme	Communication Cost	Assumption	Local Adaptive Clipping	Global Adaptive Clipping
DP-FedAvg [19]	$\mathcal{O}\left(\sqrt{\frac{L\Delta(\sigma_l^2+\sigma_g^2)}{SKR}}+\frac{L\Delta}{R}+\frac{\sigma^2C^2}{S^2}\right)$	Not Overcome Data Heterogeneity	No	No
DP-FedAvg-LS [20]	$\mathcal{O}\left(\sqrt{\frac{L\Delta(\sigma_l^2+\sigma_g^2)}{SKR}}+\frac{L\Delta}{R}+\frac{\sigma^2C^2}{S^2}\right)$	Not Overcome Data Heterogeneity	Yes	No
DP-FedSAM [17]	$\mathcal{O}\left(\sqrt{\frac{L\Delta(\sigma_l^2+\sigma_g^2)}{SKR}}+\frac{L\Delta}{R}+\frac{\sigma^2C^2}{S^2}\right)$	Not Overcome Data Heterogeneity	Yes	No
DP-SCAFFOLD [21]	$\mathcal{O}\left(2\sqrt{\frac{L\Delta\sigma_l^2}{SKR}}+2\frac{L\Delta}{R}+2\frac{\sigma^2C^2}{S^2}\right)$	Overcome Data Heterogeneity	No	No
FedAdaClip(Ours)	$\mathcal{O}\left(\sqrt{\frac{L\Delta\sigma_l^2}{SKR}}+\frac{L\Delta}{R}+\frac{\sigma^2C^2}{S^2}\right)$	Overcome Data Heterogeneity	Yes	Yes

if there exists $\lambda \in (0,1)$ such that $\sigma^2/4\mathcal{L}^2 \geq \frac{2}{3}$ and $\alpha - 1 \leq \frac{\sigma^2}{6\mathcal{L}^2}\log(1/(\tau\alpha(1+\sigma^2/4\mathcal{L}^2)))$, where $\alpha = \log(1/\delta)/((1-\lambda)\epsilon)+1$.

B. Noise Minimization Analysis

Theorem 2 (Noise Minimization)**.** *To minimize the Gaussian noise $\left\|\boldsymbol{b}^t\right\|\sigma$, the squared norm of the transformed gradient $\mathbb{E}\left\|\boldsymbol{\tau}^t\right\|^2$ must be kept bounded. This minimization (or bounded gradient norm) is achieved only when the following Gaussian noise is introduced:*

$$\boldsymbol{a}_i^t = m_i^t \quad \textit{and} \quad \boldsymbol{b}_i^t = \sqrt{\vartheta_i^t \Gamma/\zeta} \tag{19}$$

where $i \in [1,d]$ denotes the coordinate index, $\vartheta_i^t = \sqrt{\mathbf{V}\left(\boldsymbol{g}_i^t\right)}$ represents the square root of the gradient variance computed coordinate-wise, and $\Gamma = \sum_{j=1}^d \vartheta_i^t$ represents the sum of all coordinates ϑ for the current client.

1) Noise Minimization Analysis: Theorem 1 provides a rigorous formulation for minimizing the magnitude of Gaussian noise in the differentially private federated optimization setting. Specifically, the goal is to bound or minimize the term $\left\|\boldsymbol{b}^t\right\|\sigma$, which represents the scale of noise injected to preserve privacy. This term is directly influenced by the transformation parameters used in gradient preprocessing before noise addition. To achieve noise minimization, the theorem introduces a per-coordinate adaptive transformation:

$$\boldsymbol{a}_i^t = \boldsymbol{m}_i^t \quad \text{and} \quad \boldsymbol{b}_i^t = \sqrt{\vartheta_i^t \Gamma/\zeta} \tag{20}$$

where $\boldsymbol{a}_i^t$ is the mean offset of the gradient transformation, matched to the local running mean $\boldsymbol{m}_i^t$; $\boldsymbol{b}_i^t$ is the per-coordinate scaling factor used to normalize the gradient before noise addition; $\vartheta_i^t = \sqrt{\mathbf{V}(\boldsymbol{g}_i^t)}$ is the square root of the variance of the i-th coordinate of the gradient; $\Gamma = \sum_{j=1}^d \vartheta_j^t$ is the aggregate variance magnitude across all d coordinates; ζ is a normalization constant that controls the total clipping budget.

Interpretation This formulation suggests that optimal noise calibration can be achieved by proportionally scaling each coordinate according to its local variance. Coordinates with larger variance receive higher scaling factors, ensuring that the transformed gradients remain balanced across dimensions. The use of variance-aware scaling helps to:

1) Normalize gradients such that their transformed norm $\mathbb{E}\left\|\boldsymbol{\tau}^t\right\|^2$ remains bounded;
2) Prevent over-amplification of low-variance dimensions (which could lead to under-clipping or excessive noise);
3) Reduce total noise magnitude $\left\|\boldsymbol{b}^t\right\|\sigma$ under a fixed privacy budget, thus improving model utility.

Implication The theorem offers a principled method for per-coordinate noise scaling in DP-FL, forming the theoretical basis for the adaptive clipping strategy in `FedAdaClip`. It ensures that privacy-preserving noise is distributed efficiently across gradient dimensions while keeping the overall variance of the clipped gradients under control. This balance is critical for both convergence and accuracy in non-IID federated learning settings.

C. Convergence Analysis

In this section, we provide the convergence guarantees for our proposed `FedAdaClip`. First, we introduce some common assumptions for our theoretical analysis.

Assumption 1. *Each $f_i(\boldsymbol{x})$ is L-smooth, if*

$$\|\nabla f_i(\boldsymbol{x}) - \nabla f_i(\boldsymbol{y})\| \leq L\|\boldsymbol{x}-\boldsymbol{y}\|, \quad \textit{for any } \boldsymbol{x},\boldsymbol{y}. \tag{21}$$

Assumption 2. *$g_i(\boldsymbol{x}) = \nabla f_i(\boldsymbol{x})$ is the unbiased stochastic mini-batch gradient of f_i with bounded variance. The variance of $g_i(\boldsymbol{x})$ satisfies*

$$\mathbb{E}[\|g_i(\boldsymbol{x}) - \nabla f_i(\boldsymbol{x})\| \leq \sigma_l^2, \quad \textit{for any } i,\boldsymbol{x}, \tag{22}$$

Theorem 3 (Convergence for non-convex functions)**.** *Under Assumption 1-2, if we set $\boldsymbol{g}^0 = 0, \beta, \gamma$, and η, `FedAdaClip` enjoys:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla f\left(\boldsymbol{x}_t\right)\right\|^2\right] \lesssim \sqrt{\frac{L\Delta(\sigma^2+L^2\sigma^2)}{NKT}} + \frac{L\Delta}{T}, \tag{23}$$

where $\Delta \leq f\left(\boldsymbol{x}^0\right) - \min_x f(\boldsymbol{x})$ and $\lesssim$ absorbs numeric numbers.

Key Insights The convergence bound

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\left\|\nabla f\left(\boldsymbol{x}_t\right)\right\|^2\right] \lesssim \sqrt{\frac{L\Delta(\sigma^2+L^2\sigma^2)}{NKT}} + \frac{L\Delta}{T}$$

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indicates that the expected gradient norm diminishes at a rate of $\mathcal{O}(1/\sqrt{NT})$, aligning with the best-known rates of centralized and DP-FL methods. The bound also reflects how differential privacy noise σ^2 and local smoothness $\mathcal{L}$ affect convergence, affirming that `FedAdaClip` maintains stability and performance under privacy constraints. Furthermore, the clipping-adaptive design—anchored by EMA and tunable via parameter β—offers robustness to heterogeneity and noisy dynamics. As shown in Table I, we compare FedAdaClip with existing DP-FL algorithms in terms of theoretical convergence rate, ability to overcome data heterogeneity, and whether local/global adaptive clipping is applied. DP-FedAvg and its variants (e.g., DP-FedAvg-LS and DP-FedSAM) achieve the same order of communication complexity as FedAdaClip, but they either rely on fixed clipping thresholds or only support local clipping, thus failing to effectively address data heterogeneity. DP-SCAFFOLD alleviates the heterogeneity issue by control variates, yet it does not adopt adaptive clipping. In contrast, FedAdaClip incorporates both local and global adaptive clipping mechanisms, which not only preserves the same theoretical convergence rate but also enhances robustness against heterogeneous data. This joint design is the key reason why FedAdaClip achieves more stable convergence curves and higher accuracy in our experiments.

Implication The result confirms that `FedAdaClip` can achieve reliable convergence in federated settings without requiring strong assumptions on client similarity or convexity. This distinguishes our method from prior work that either suffers from degraded convergence under noise or fails to generalize in heterogeneous data environments.

VI. Experimental Analysis

A. Experimental Setup

Models and Datasets. We adopt a deep neural network architecture consistent with prior schemes to evaluate `FedAdaClip` on image classification tasks across three benchmark datasets: MNIST [28], CIFAR-10 [29], and CIFAR-100 [29]. Specifically, we employ a convolutional neural network (CNN) comprising two convolutional layers and two fully connected layers. Each convolutional layer is followed by a ReLU activation and a max-pooling operation. The convolutional kernels are of size 5×5 with a stride of 1 and zero-padding, while max-pooling is performed with 2×2 kernels and a stride of 2.

As our goal is to evaluate the effectiveness of the optimization algorithm rather than explore novel network architectures, standard CNNs are sufficient for our experiments. To simulate data heterogeneity, we use a Dirichlet distribution [30] to partition the dataset across clients, where each client's local dataset follows a Dirichlet distribution with concentration parameter ρ and uniform prior probability q.

Dirichlet Subsampling. To generate non-IID client datasets, we adopt Dirichlet subsampling, where the dataset of each client D_i is drawn from $\mathcal{D}_i \sim Dir(\rho \cdot q)$. Here, $\rho > 0$ is the concentration parameter that controls the degree of data heterogeneity, and q represents a uniform prior distribution. A larger ρ results in more IID data distributions among clients, whereas a smaller ρ induces higher statistical heterogeneity.

Implementation Details. We compare our proposed `FedAdaClip` with recent federated learning baselines, including DP-FedAvg [19], DP-SCAFFOLD [21], DP-FedAvg-LS [20], and DP-FedSAM [17]. The total number of clients is set to $N = 100$, with a mini-batch size of $B = 100$. The clipping threshold is fixed at $C = 10.0$, and the client sampling rate is set to $\gamma = 0.1$. We vary the local learning rate η from 0.025 to 0.25, and the global learning rate η_g from 1.0 to 2.5. Local training is performed for $E \in \{1, 5\}$ epochs, where $E = 1$ corresponds to synchronous updates, and $E = 5$ allows for multiple local updates, potentially causing client drift. All experiments are implemented in PyTorch 1.13.0 and executed on two NVIDIA RTX 3090 GPUs.

B. Main Results

We evaluate the performance of `FedAdaClip` under a fixed privacy budget $\epsilon = 3$ across three benchmark datasets: MNIST, CIFAR-10, and CIFAR-100, each with varying local update steps $\rho \in \{0.1, 0.6, 1.0, 10.0\}$. Tables II–IV report the average test accuracy (%) under different data heterogeneity levels.

a) Performance on MNIST: As shown in Table II, `FedAdaClip` consistently outperforms all baseline algorithms across all values of ρ. Compared to DP-FedSAM and DP-FedAvg-LS, it achieves up to +3.27% and +2.16% improvements, respectively, while maintaining strong privacy guarantees. The improvements become more prominent under high heterogeneity (low ρ), demonstrating `FedAdaClip`'s robustness to non-IID data. `FedAdaClip` demonstrates consistent advantages across all local steps ρ.

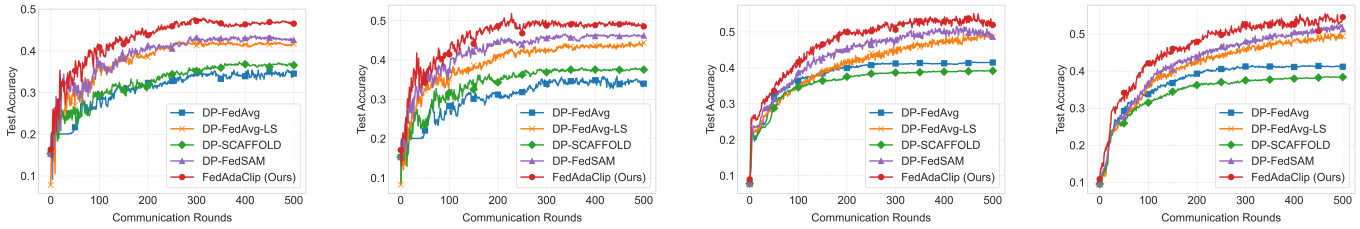
TABLE II: Test Accuracy (%) on MNIST under privacy budget $\epsilon = 3$.

Scheme	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$
DP-FedAvg [19]	74.96	84.81	88.85	91.95
DP-SCAFFOLD [21]	90.65	91.49	90.57	90.55
DP-FedSAM [17]	94.14	95.18	95.93	96.29
DP-FedAvg-LS [20]	82.87	95.14	95.74	96.19
`FedAdaClip` (Ours)	**95.34**	**98.41**	**98.39**	**98.75**

b) Performance on CIFAR-10: Table III indicates that although `FedAdaClip` and DP-FedSAM achieve similar performance, `FedAdaClip` eliminates the need for computationally expensive smoothing steps (as required in FedSAM). Notably, it achieves +5.7% accuracy improvement over DP-FedAvg-LS at $\rho = 1.0$, highlighting its advantage under moderate heterogeneity. `FedAdaClip` achieves superior performance with reduced overhead.

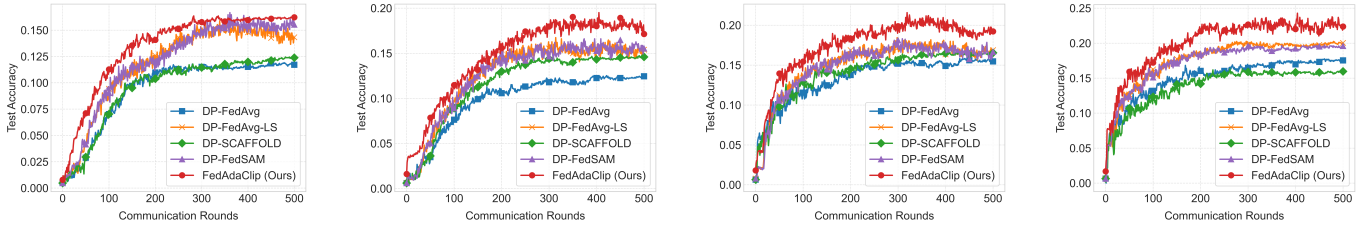
c) Performance on CIFAR-100: As shown in Table IV, `FedAdaClip` delivers consistent improvements on CIFAR-100 as well. Despite the relatively low baseline performance due to model capacity limitations, our method surpasses both DP-FedSAM and DP-FedAvg-LS by +1–2% across all ρ settings. This confirms that adaptive clipping helps stabilize

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(a) $\rho = 0.1$ (b) $\rho = 0.6$ (c) $\rho = 1.0$ (d) $\rho = 10.0$

Fig. 2: The impact of data heterogeneity on CIFAR-10.



(a) $\rho = 0.1$ (b) $\rho = 0.6$ (c) $\rho = 1.0$ (d) $\rho = 10.0$

Fig. 3: The impact of data heterogeneity on CIFAR-100.

TABLE III: Test Accuracy (%) on CIFAR-10 under privacy budget $\epsilon = 3$.

Scheme	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$
DP-FedAvg [19]	35.71	35.86	41.54	41.51
DP-SCAFFOLD [21]	37.34	38.03	39.24	40.44
DP-FedSAM [17]	42.91	46.63	50.28	51.51
DP-FedAvg-LS [20]	42.32	44.60	49.62	50.64
`FedAdaClip` (Ours)	**48.01**	**49.12**	**54.31**	**52.37**

training even on more complex datasets. `FedAdaClip` improves performance across all heterogeneity levels.

TABLE IV: Test Accuracy (%) on CIFAR-100 under privacy budget $\epsilon = 3$.

Scheme	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$
DP-FedAvg [19]	11.23	12.34	15.34	17.47
DP-SCAFFOLD [21]	12.15	14.33	16.41	16.52
DP-FedSAM [17]	16.35	16.49	18.51	19.44
DP-FedAvg-LS [20]	15.56	16.64	18.63	19.89
`FedAdaClip` (Ours)	**17.67**	**18.20**	**19.92**	**20.88**

Across all benchmarks, `FedAdaClip` demonstrates superior generalization under strong privacy constraints and varying degrees of client heterogeneity. Its robustness stems from the dual-phase clipping design, which balances stability (via quantile-based thresholding) and adaptivity (via EMA and gradient-aware scaling). Unlike DP-FedSAM, it avoids costly preprocessing, making it more efficient and scalable in practice.

C. Impact of Data Heterogeneity

Fig. 2 displays the training curves for various algorithms with different data heterogeneity (IID settings) on CIFAR-10. The sub-figures demonstrate that `FedAdaClip` consistently achieves superior convergence speed and performance across diverse data distributions. Specifically, `FedAdaClip` yields faster convergence and higher accuracy compared to other algorithms, regardless of whether the data is IID or non-IID. This highlights the effectiveness of `FedAdaClip` in handling data heterogeneity while maintaining robust performance.

Fig. 3 illustrates the training loss curves under the same heterogeneous settings. We observe that FedAdaClip converges faster and exhibits more stable loss reduction compared to other methods. This confirms that the proposed global adaptive clipping not only improves the final accuracy but also accelerates convergence by mitigating the adverse effects of non-IID data distributions.

D. Impact of Privacy Budget

We study the performance of `FedAdaClip` under varying privacy budgets $\epsilon \in 1, 3, 5, 7, 9$ on both CIFAR-10 and CIFAR-100 datasets, as shown in TableVI. As expected, larger values of ϵ (weaker privacy guarantees) lead to higher test accuracy and lower training loss. On CIFAR-10, performance improves from 45.52% to 59.80%, while CIFAR-100 shows a more modest improvement due to its higher complexity. The model benefits from looser privacy constraints. This demonstrates that `FedAdaClip` maintains strong utility across a wide privacy spectrum, making it a flexible choice for privacy-preserving federated learning.

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TABLE V: Comparison of accuracy with FedMAC and AdaClip under different ρ.

Scheme	MNIST				CIFAR-10				CIFAR-100			
	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$	$\rho = 0.1$	$\rho = 0.6$	$\rho = 1.0$	$\rho = 10.0$
FedMAC	93.59	94.82	94.67	95.13	44.25	44.65	50.12	50.23	14.23	15.66	16.52	17.65
AdaClip	93.61	94.47	94.76	95.32	45.25	77.32	51.22	51.66	14.56	16.28	17.62	18.12
FedAdaClip (Ours)	**95.34**	**98.41**	**98.39**	**98.75**	**48.01**	**49.12**	**54.31**	**52.37**	**17.67**	**18.20**	**19.92**	**20.88**

TABLE VI: FedAdaClip performance on CIFAR-10 and CIFAR-100 under different privacy budgets ϵ.

ϵ	CIFAR-10		CIFAR-100	
	Test Acc. (%)	Train Loss	Test Acc. (%)	Train Loss
1	45.52	2.03	16.48	4.26
3	54.31	1.32	19.92	3.88
5	57.48	1.19	21.35	3.45
7	58.42	1.14	21.73	3.34
9	**59.80**	**1.06**	**22.47**	**3.18**

E. Impact of Quantile Parameter γ

We conduct ablation studies on the quantile parameter γ to evaluate its impact on performance. γ controls the clipping aggressiveness. As shown in Tables VII, setting $\gamma = 0.8$ achieves the best trade-off between accuracy and loss on both CIFAR-10 and CIFAR-100 datasets. A too-small γ clips too many gradients, reducing learning efficiency, while a too-large γ under-clips, leading to noise-dominated updates. This highlights the importance of tuning the clipping quantile in differentially private FL scenarios.

TABLE VII: Impact of γ on CIFAR-10 and CIFAR-100.

γ	CIFAR-10		CIFAR-100	
	Test Acc. (%)	Train Loss	Test Acc. (%)	Train Loss
0.50	51.32	1.78	17.43	4.26
0.60	52.65	1.58	18.01	4.03
0.70	53.91	1.41	18.52	3.88
0.80	**54.31**	**1.32**	**19.92**	**3.68**
0.90	52.64	1.38	19.33	3.89
0.95	51.09	1.54	18.48	4.07

F. Compared with AdaClip

AdaClip [14] and FedMAC [15] are single-machine adaptive clipping methods. We implemented FedMAC and AdaClip in the federated learning framework and used them as comparative schemes for adaptive threshold clipping, and their performance is validated through image classification tasks.

Table V reports the image classification accuracy (%) on MNIST, CIFAR-10 and CIFAR-100 under different heterogeneity levels ρ. It can be observed that our proposed scheme, FedAdaClip, consistently outperforms both FedMAC and AdaClip across all settings. Notably, under extreme heterogeneity ($\rho = 10.0$), FedAdaClip achieves a significant performance gain, reaching 98.75%, which is approximately 3.6% higher than the best baseline (FedMAC at 95.13%). Even under mild heterogeneity ($\rho = 0.6$ and 1.0), our scheme maintains a clear advantage, indicating superior adaptability to non-IID data. These results demonstrate the robustness and effectiveness of FedAdaClip in heterogeneous federated learning scenarios.

On CIFAR-10, our scheme yields clear improvements over FedMAC and AdaClip, particularly under highly heterogeneous data distributions (e.g., $\rho = 0.1$), where FedAdaClip surpasses AdaClip by more than 2.7%. Similarly, on CIFAR-100, the advantage of FedAdaClip becomes more pronounced: at $\rho = 0.1$, FedAdaClip improves accuracy by over 3% compared to AdaClip, and the margin remains significant across all ρ. These results demonstrate that the proposed global-and-local adaptive clipping strategy provides robust performance gains on both relatively simple (CIFAR-10) and more challenging (CIFAR-100) benchmarks, effectively mitigating the negative impact of data heterogeneity.

G. Ablation Experiments

To evaluate the contribution of each component in FedAdaClip, we conduct ablation experiments on CIFAR-10 under the same privacy budget ($\epsilon = 3$) and heterogeneity level ($\rho = 1.0$). We isolate the contribution of each component in FedAdaClip. As shown in Table VIII, removing the quantile initialization or the gradient-aware adjustment significantly degrades performance, indicating their complementary roles. Moreover, using a fixed threshold or discarding EMA smoothing leads to higher loss and lower accuracy, demonstrating the importance of adaptivity and temporal stability. The full FedAdaClip configuration achieves the best result, validating the effectiveness of our design.

TABLE VIII: Ablation experiments on CIFAR-10 under $\epsilon = 3$ and $\rho = 1.0$.

Method Variant	Test Acc. (%)	Train Loss
w/o Quantile Init	51.48	1.55
w/o Gradient-aware	50.77	1.61
w/o EMA	48.94	1.73
Fixed Clipping ($C = 1.0$)	47.66	1.85
FedAdaClip (Ours)	**54.31**	1.32

VII. Conclusion

This paper introduces FedAdaClip, a differentially private federated learning scheme designed to address the challenge of setting gradient clipping thresholds in existing DP-FL

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framework. FedAdaClip combines quantile-based clipping strategies with gradient-aware strategies, dynamically adjusting the clipping threshold using exponential moving average. FedAdaClip not only adheres to strict differential privacy guarantees but also enhances the convergence speed and accuracy of the model. Theoretical analysis shows that FedAdaClip guarantees convergence for non-convex objectives and minimizes noise. Experimental results further validate the effectiveness of FedAdaClip, which outperforms existing fixed-threshold clipping algorithms on multiple standard datasets, especially in handling highly heterogeneous data. Additionally, FedAdaClip accelerates convergence in the early stages of training and effectively reduces the cumulative impact of noise in the later stages, ultimately yielding a superior model performance. These results indicate that FedAdaClip provides an effective solution to the gradient clipping problem in differentially private federated learning.

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